



NORMANHURST BOYS HIGH SCHOOL

MATHEMATICS EXTENSION 1

2020 Year 12 Course Assessment Task 4 (Trial Examination)

Thursday August 27, 2020

General instructions

- Working time – 2 hours.
(plus 10 minutes reading time)
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- NESA approved calculators may be used.
- Attempt **all** questions.
- At the conclusion of the examination, bundle the booklets used in the correct order within this paper and hand to examination supervisors.

SECTION I

- Mark your answers on the answer grid provided (on page 11)

SECTION II

- Commence each new question on a new booklet. Write on both sides of the paper.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.

NESA STUDENT #: # BOOKLETS USED:

Class (please ✓)

☐ 12MXX.1 – Ms Ham

☐ 12MAX.3 – Mr Sekaran

☐ 12MAX.5 – Mr Siu

☐ 12MXX.2 – Mr Lam

☐ 12MAX.4 – Ms Bhamra

Marker's use only.

QUESTION	1-10	11	12	13	14	Total	%
MARKS	$\overline{10}$	$\overline{17}$	$\overline{15}$	$\overline{14}$	$\overline{14}$	$\overline{70}$	

Section I

10 marks

Attempt Question 1 to 10

Allow approximately 15 minutes for this section

Mark your answers on the answer grid provided (labelled as page 11).

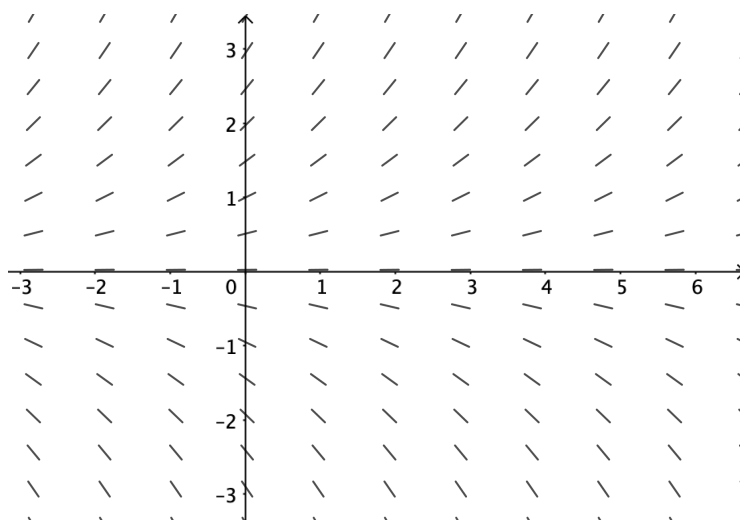
Questions

Marks

1. What is the exact value of $\sec\left(-\frac{5\pi}{3}\right)$? 1

(A) $\frac{1}{2}$ (B) $-\frac{1}{2}$ (C) -2 (D) 2

2. Consider the slope field shown below. 1



What is a possible differential equation for the slope field?

(A) $\frac{dy}{dx} = -\frac{x}{2}$ (B) $\frac{dy}{dx} = \frac{x}{2}$ (C) $\frac{dy}{dx} = -\frac{y}{2}$ (D) $\frac{dy}{dx} = \frac{y}{2}$

3. What is the smallest positive x value for which $\sin x + \cos x$ is at its maximum? 1

(A) $\frac{\pi}{2}$ (B) $\frac{\pi}{4}$ (C) π (D) $\frac{\pi}{3}$

4. Which one of the following transformations is **not** required to obtain $g(x) = -1 + \frac{1}{2}f(3 - 2x)$ from $f(x)$? 1

- (A) A reflection in the y axis
- (B) A reflection in the x axis
- (C) A translation parallel to the x axis
- (D) A compression in the y direction

5. Which one of the following is the correct description of the asymptote(s) of 1

$$y = \frac{1}{x^2 + 7x - 8}$$

- (A) exactly one straight line asymptote.
- (B) exactly two straight line asymptotes.
- (C) exactly three straight line asymptotes.
- (D) $x = -1$ and $x = 8$ as its vertical asymptotes.

6. Consider the polynomial $P(x) = x^3 + x^2 + cx - 10$. It is known that two of its zeros are equal in magnitude but opposite in sign. 1

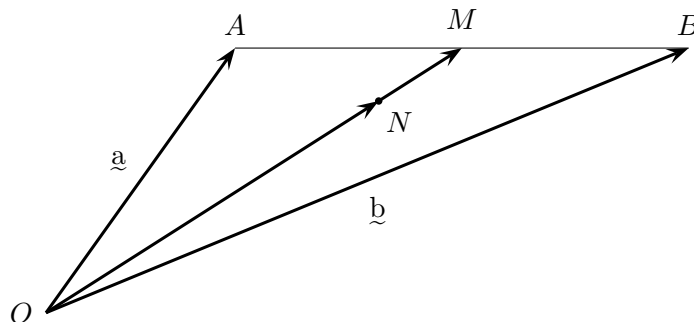
What is the value of c ?

- (A) $\sqrt{10}$ (B) 10 (C) -10 (D) $-\sqrt{10}$

7. Which expression is equal to $\int \cos 3x \cos 2x \, dx$? 1

- (A) $\frac{1}{10} \sin 5x + \frac{1}{2} \sin x + c$ (C) $\frac{1}{10} \cos 5x + \frac{1}{2} \cos x + c$
- (B) $\frac{1}{2}(\cos 5x + \cos x) + c$ (D) $\frac{1}{2}(\sin 5x + \sin x) + c$

8. In the following diagram, M is the midpoint of line segment AB , and $\overrightarrow{ON} = \frac{4}{5}\overrightarrow{OM}$. 1
 It is given that $\overrightarrow{OA} = \underline{a}$ and $\overrightarrow{OB} = \underline{b}$.

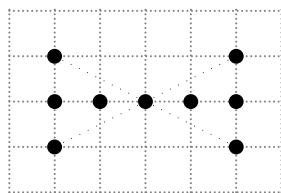


Which of the following expressions correctly relates $\overrightarrow{NM}$ with $\underline{a}$ and $\underline{b}$?

- (A) $\overrightarrow{NM} = \frac{4}{5}(\underline{a} - \underline{b})$ (C) $\overrightarrow{NM} = \frac{1}{10}(\underline{a} + \underline{b})$
 (B) $\overrightarrow{NM} = \frac{2}{5}(\underline{a} + \underline{b})$ (D) $\overrightarrow{NM} = \frac{1}{10}(\underline{a} - \underline{b})$
9. What is the limiting sum of the following geometric series? 1

$$1 - \cos(2x) + \cos^2(2x) - \cos^3(2x) + \dots$$

- (A) $\frac{1}{2} \sec^2 x$ (C) $\frac{1}{2} \cos^2 x$
 (B) $\frac{1}{2} \operatorname{cosec}^2 x$ (D) $\frac{1}{2} \sin^2 x$
10. The diagram below shows a shape made by 9 points. 1



How many combinations of 3 points are collinear?

- (A) 10 (B) 14 (C) 12 (D) 8

Examination continues overleaf...

Section II

60 marks

Attempt Questions 11 to 14

Allow approximately 1 hours and 45 minutes for this section.

Write your answers in the writing booklets supplied. Additional writing booklets are available. Your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (17 Marks)	Commence a NEW booklet.	Marks
(a) Given the function $f(x) = \sqrt{2x+1}$ and that $f^{-1}(x)$ is the inverse function of $f(x)$, find $f^{-1}(5)$.	2	
(b) Solve for x : $\frac{x}{x+1} > 2$	3	
(c) Use the method of addition of ordinates to sketch the graph of $f(x) = e^{-x} - \frac{x}{e}$ given $e \approx 2.7$. It is also given that the x -intercept is $x = 1$. Show all important features.	3	
(d) i. Use t -formulae to show that	1	
$\frac{1 - \cos \theta}{\sin \theta} = \tan \frac{\theta}{2}.$		
ii. Hence, or otherwise, find the exact value of $\tan 15^\circ$.	1	
(e) Find the exact value of $\tan \left(2 \sin^{-1} \frac{2}{3} \right)$	2	
(f) At a soccer club a team of 13 players is to be chosen from a pool of 25 players consisting of 20 female players and 5 male players.	2	
What is the probability that the team will consist of only female players?		
(g) Find the term independent of x in the expansion of $\left(3x - \frac{1}{2x^2} \right)^9$	3	

Examination continues overleaf...

Question 12 (15 Marks)

Commence a NEW booklet.

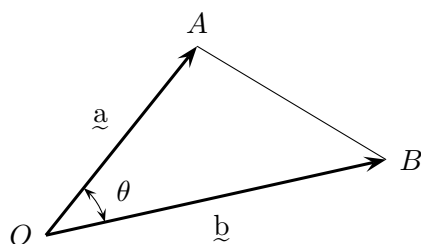
Marks

- (a) If $x = \alpha$ is a double root of a polynomial $P(x)$, show that $x = \alpha$ is also a root of $P'(x)$. **2**

Hint: Let $P(x) = (x - \alpha)^2 Q(x)$

- (b) A force, described by the vector $\mathbf{F} = \begin{pmatrix} 2 \\ 5 \end{pmatrix}$, moves a particle along the line ℓ from $A(-1, -1)$ to $B(2, -3)$. **2**
- Find the unit vector that is in the direction of $\overrightarrow{AB}$. **2**
 - Find the component of $\mathbf{F}$ in the direction of line ℓ . **2**

- (c) The diagram below shows $\triangle OAB$, where $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OB} = \mathbf{b}$. **3**



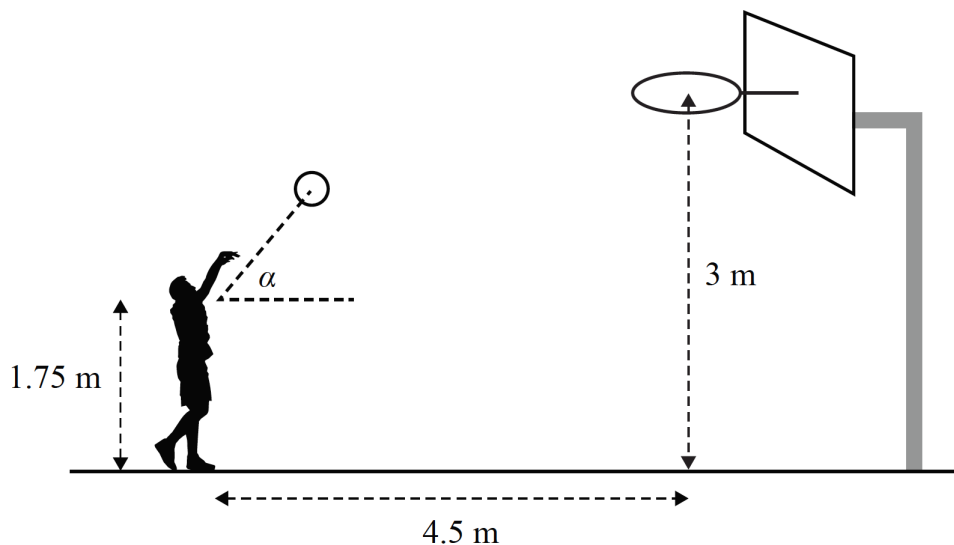
Given that θ is the angle between vectors $\mathbf{a}$ and $\mathbf{b}$, show that the area of the triangle is

$$A = \frac{1}{2} \sqrt{(\mathbf{a} \cdot \mathbf{a})(\mathbf{b} \cdot \mathbf{b}) - (\mathbf{a} \cdot \mathbf{b})^2}$$

Examination continues overleaf...

Question 12 continued from previous page...

- (d) A basketball player aims to throw a basketball through a ring, the centre of which is at a horizontal distance of 4.5 m from the point of release of the ball and 3 m above floor level. The ball is released at a height of 1.75 m above floor level, at an angle of projection α to the horizontal and at a speed of $V \text{ ms}^{-1}$. Air resistance is assumed to be negligible.



The position vector of the centre of the ball at any time t seconds, for $t \geq 0$, relative to the point of release is given by

$$\mathbf{r} = \begin{pmatrix} Vt \cos \alpha \\ Vt \sin \alpha - 5t^2 \end{pmatrix}$$

Displacement components are measured in metres. For the player's first shot at goal, $V = 7 \text{ ms}^{-1}$ and $\alpha = 45^\circ$.

- i. Find the time, in seconds, taken for the ball to reach its maximum height. **2**

Give your answer in the form $\frac{a\sqrt{b}}{c}$, where a, b and c are positive integers.

- ii. Find the maximum height, in metres, above floor level, reached by the centre of the ball. **2**

Give your answer correct to two decimal places.

- iii. Find the distance between the centre of the ball and the centre of the ring when the ball reaches its maximum height. **2**

Give your answer in metres, correct to two decimal places.

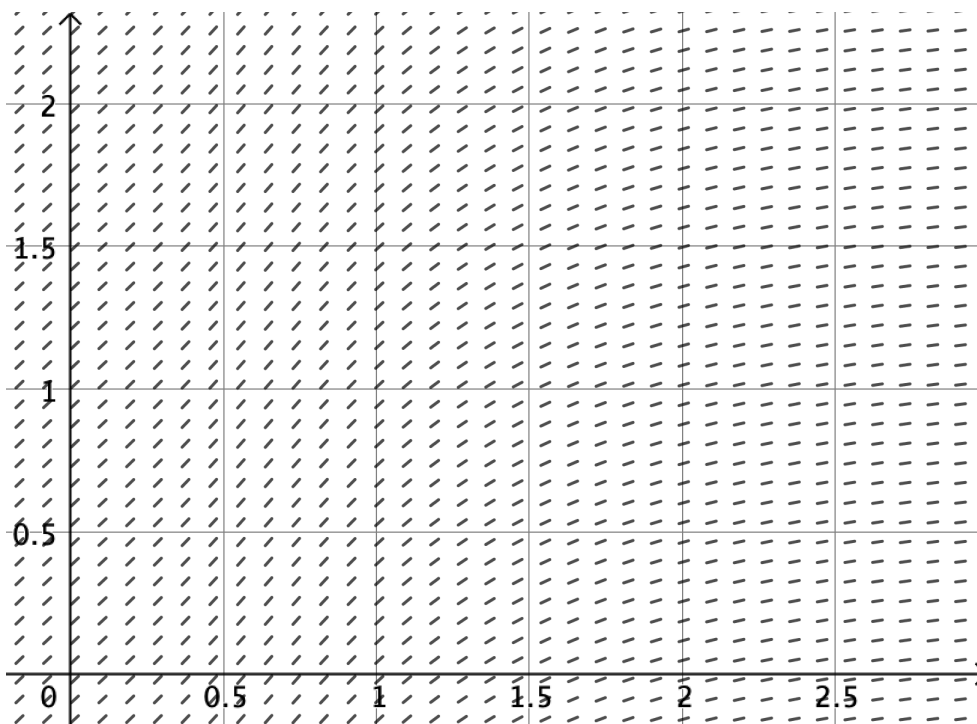
Examination continues overleaf...

Question 13 (14 Marks)	Commence a NEW booklet.	Marks
(a) Consider the differential equation $\frac{dy}{dx} - \frac{2y}{x} = 0$. Find the equation of the solution curve if it passes through $(1, -1)$.	3	
(b) Prove by mathematical induction that $3^{3n} + 2^{n+2}$ is a multiple of 5 for all positive integers n .	3	
(c) It is given that the rate of decrease of temperature of a body that is hotter than its surrounding air is proportional to the temperature difference. i.e.		
$\frac{dT}{dt} = -k(T - A)$		
where A is the air temperature, and T is the temperature of the body after t minutes		
i. Show that, if the initial temperature is I , then the following function satisfies the above differential equation.	1	
$T = A + (I - A)e^{-kt}$		
ii. A block of iron, initially at a temperature of 1500°C , is allowed to cool in the open air, where the temperature is 20°C . If it cools to 1200°C in five minutes, find the temperature of the block after one hour, correct to 3 significant figures.	2	

Examination continues overleaf...

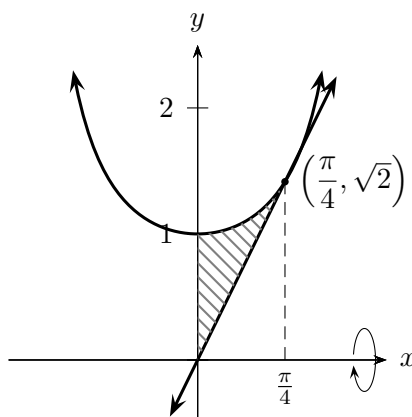
Question 13 continued from previous page...

- (d) The slope field of $\frac{dy}{dx} = \frac{1+x^2}{1+x^4}$ is shown below.



- i. It is given that the solution curve passes through $(0.5, 0.5)$. On the grid provided on page 12, sketch the solution curve on the slope field. 1
- ii. Find the approximate value of y when $x = 2$. Give your answer correct to one decimal place. 1

- (e) The diagram below shows parts of the graph of both $y = \sec x$ and $y = \frac{4\sqrt{2}}{\pi}x$. 3
- It is given that the two functions intersect at $\left(\frac{\pi}{4}, \sqrt{2}\right)$ for the first time in the first quadrant.



The area bounded by the curve $y = \sec x$, $y = \frac{4\sqrt{2}}{\pi}x$ and the y axis is rotated about the x axis. Find the volume of the solid of revolution.

Examination continues overleaf...

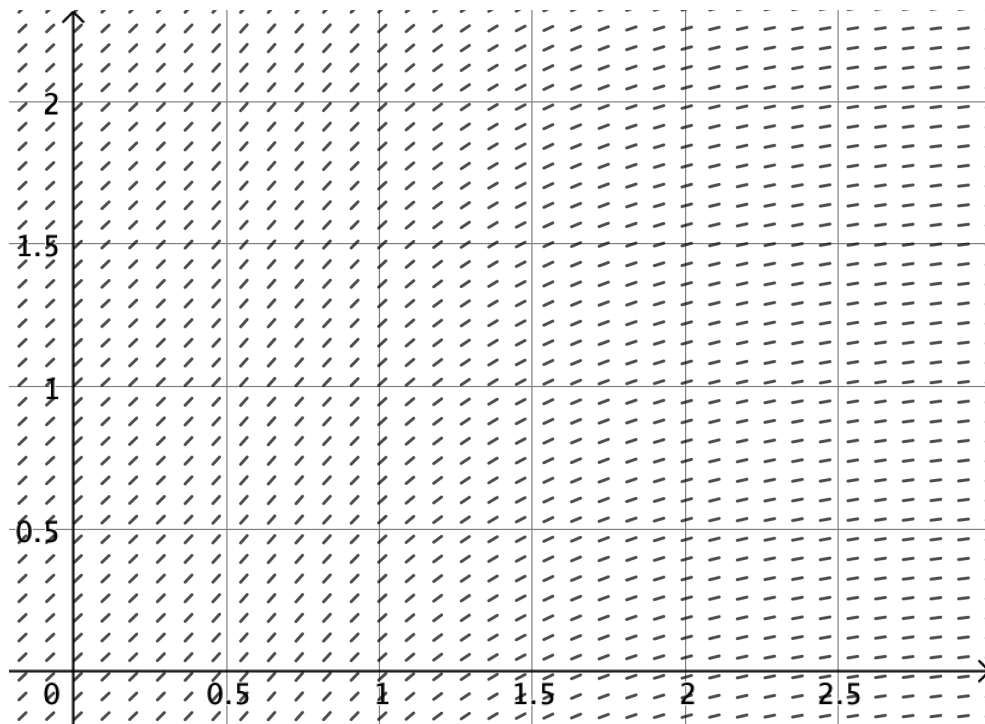
- Question 14** (14 Marks) Commence a NEW booklet. **Marks**
- (a) Use the substitution $u = 1 + \sqrt{x}$ to evaluate **4**
- $$\int_1^9 \frac{\sqrt{x} + 2}{\sqrt{x}\sqrt{1 + \sqrt{x}}} dx$$
- , expressing the answer in the form $\sqrt{n}$ where n is a positive integer.
- (b) The volume of water in a tank is given by $V = \frac{5\pi}{6} - \cos^{-1}\left(h - \frac{\sqrt{3}}{2}\right)$, where V is measured in m^3 and h is depth of the water in the tank in metres.
- i. Find the domain and range of $V = \frac{5\pi}{6} - \cos^{-1}\left(h - \frac{\sqrt{3}}{2}\right)$. **2**
- ii. Sketch the graph of $V = \frac{5\pi}{6} - \cos^{-1}\left(h - \frac{\sqrt{3}}{2}\right)$, showing all important features. **3**
- From time $t = 0$, where t is measured in minutes, water is pumped into the empty tank at a constant rate of 50 litres per minute.
(Note: $1\text{ m}^3 = 1\,000\text{ L}$)
- iii. Find the expression of V in terms of t . **1**
- iv. Find the exact time in minutes required to reach the maximum volume of water. **1**
- v. Find the rate of increase of the depth when the water in the tank is $\sqrt{3}$ metres deep. **3**

End of paper.

Slope field

Question 13

- (d) i. It is given that the solution curve passes through $(0.5, 0.5)$. Sketch the solution curve on the slope field below. See Question 13((d))i on page 9. 1



Sample Band 6 Responses

Section I

1. (D) 2. (D) 3. (B) 4. (B) 5. (C)
6. (C) 7. (A) 8. (C) 9. (A) 10. (B)

Section II

Question 11

(a) (2 marks)

- ✓ [1] for an equation involving $f(x)$
✓ [1] for final answer

$$\begin{aligned} 5 &= \sqrt{2x+1} \\ \text{Squaring both sides,} \\ 25 &= 2x+1 \\ 2x &= 24 \\ x &= 12 \\ \therefore f^{-1}(5) &= 12 \end{aligned}$$

(b) (3 marks)

- ✓ [1] for multiplying both sides by $(x+1)^2$
✓ [1] for the correct quadratic inequality
✓ [1] for final answer

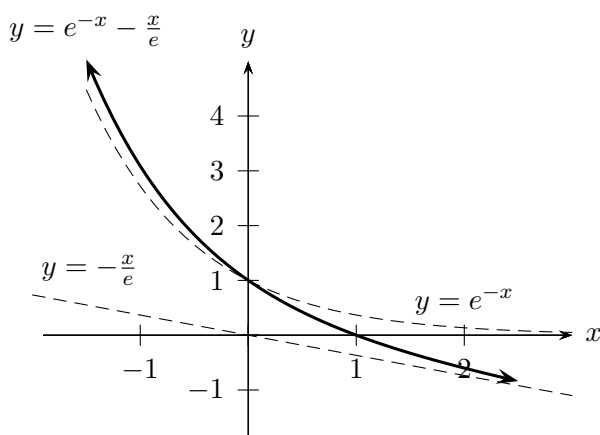
$$\frac{x}{x+1} > 2$$

Multiply both sides by $(x+1)^2$

$$\begin{aligned} x(x+1) &> 2(x+1)^2 \\ 2(x+1)^2 - x(x+1) &< 0 \\ (x+1)(2(x+1) - x) &< 0 \\ (x+1)(x+2) &< 0 \\ \therefore -2 < x < -1 \end{aligned}$$

(c) (3 marks)

- ✓ [1] for shape
✓ [1] for y -intercept
✓ [1] for approaching $y = -\frac{x}{e}$ as $x \rightarrow \infty$



(d) i. (1 mark)

$$\begin{aligned} \text{LHS} &= \frac{1 - \cos \theta}{\sin \theta} \\ &= \frac{1 - \left(\frac{1-t^2}{1+t^2}\right)}{\frac{2t}{1+t^2}} \\ &= \frac{1+t^2-1+t^2}{2t} \\ &= \frac{2t^2}{2t} \\ &= t \\ &= \text{RHS} \end{aligned}$$

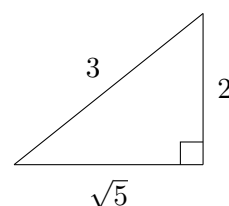
ii. (1 mark)

$$\begin{aligned} \text{Let } \theta &= 30^\circ \\ \tan 15^\circ &= \frac{1 - \cos 30^\circ}{\sin 30^\circ} \\ &= \frac{1 - \frac{\sqrt{3}}{2}}{\frac{1}{2}} \\ &= 2 - \sqrt{3} \end{aligned}$$

(e) (2 marks)

- ✓ [1] for $\tan A$
✓ [1] for final answer

$$\begin{aligned} \text{Let } A &= \sin^{-1} \frac{2}{3} \\ \sin A &= \frac{2}{3} \end{aligned}$$



(b) i. (2 marks)

$$\begin{aligned}\tan 2A &= \frac{2 \tan A}{1 - \tan^2 A} \\ &= \frac{2 \times \frac{2}{\sqrt{5}}}{1 - \frac{4}{5}} \\ &= \frac{4}{\sqrt{5}} \times 5 \\ &= 4\sqrt{5}\end{aligned}$$

- ✓ [1] for $\overrightarrow{AB}$
 ✓ [1] for final answer

(f) (2 marks)

- ✓ [1] for ${}^{20}C_{13}$ or ${}^{25}C_{13}$
 ✓ [1] for final answer

$$\frac{{}^{20}C_{13}}{{}^{25}C_{13}} = \frac{12}{805}$$

$$\overrightarrow{AB} = \begin{pmatrix} 3 \\ -2 \end{pmatrix}$$

$$|\overrightarrow{AB}| = \sqrt{9 + 4} = \sqrt{13}$$

$$\therefore \text{unit vector is } \frac{1}{\sqrt{13}} \begin{pmatrix} 3 \\ -2 \end{pmatrix}$$

(g) (3 marks)

- ✓ [1] for use of binomial theorem
 ✓ [1] for value of k
 ✓ [1] for the correct term

$$\begin{aligned}\left(3x - \frac{1}{2x^2}\right)^9 &= \sum_{k=0}^9 \binom{9}{k} (3x)^{9-k} \left(-\frac{1}{2x^2}\right)^k \\ &= \sum_{k=0}^9 \binom{9}{k} 3^{9-k} \left(-\frac{1}{2}\right)^k x^{9-k} \times x^{-2k} \\ &= \sum_{k=0}^9 \binom{9}{k} 3^{9-k} \left(-\frac{1}{2}\right)^k x^{9-3k}\end{aligned}$$

The term independent of x is when $k = 3$.

$$\therefore \binom{9}{3} 3^6 \left(-\frac{1}{2}\right)^3 = -7654.5 \text{ or } -\frac{15309}{2}$$

Question 12

(a) (2 marks)

- ✓ [1] for $P'(x)$
 ✓ [1] for showing α is a zero of $P'(x)$ by substitution

(c) (3 marks)

$$\begin{aligned}P(x) &= (x - \alpha)^2 Q(x) \\ P'(x) &= 2(x - \alpha)Q(x) + (x - \alpha)^2 Q'(x) \\ P'(\alpha) &= 0 + 0 \\ &= 0 \\ \therefore x = \alpha &\text{ is a zero of } P'(x).\end{aligned}$$

ii. (2 marks)

- ✓ [1] for use of projection formula
 ✓ [1] for final answer

Applying the projection of $\mathbb{F}$ on to $\overrightarrow{AB}$, such that

$$\begin{aligned}\text{proj}_{\mathbb{F}} \overrightarrow{AB} &= \frac{\mathbb{F} \cdot \overrightarrow{AB}}{|\overrightarrow{AB}|^2} \overrightarrow{AB} \\ &= \frac{\begin{pmatrix} 2 \\ 5 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -2 \end{pmatrix}}{13} \begin{pmatrix} 3 \\ -2 \end{pmatrix} \\ &= \frac{6 - 10}{13} \begin{pmatrix} 3 \\ -2 \end{pmatrix} \\ &= -\frac{4}{13} \begin{pmatrix} 3 \\ -2 \end{pmatrix}\end{aligned}$$

- ✓ [1] for A^2 in terms of $\cos^2 \theta$
 ✓ [1] for substituting $\cos^2 \theta = \frac{(\underline{a} \cdot \underline{b})^2}{|\underline{a}|^2 |\underline{b}|^2}$ into A^2
 ✓ [1] for final result

$$\begin{aligned}
 A &= \frac{1}{2} |\underline{a}| |\underline{b}| \sin \theta \\
 A^2 &= \frac{1}{4} |\underline{a}|^2 |\underline{b}|^2 \sin^2 \theta \\
 &= \frac{1}{4} |\underline{a}|^2 |\underline{b}|^2 (1 - \cos^2 \theta) \\
 \therefore \cos \theta &= \frac{(\underline{a} \cdot \underline{b})}{|\underline{a}| |\underline{b}|}, \\
 \cos^2 \theta &= \frac{(\underline{a} \cdot \underline{b})^2}{|\underline{a}|^2 |\underline{b}|^2} \\
 A^2 &= \frac{1}{4} |\underline{a}|^2 |\underline{b}|^2 \left(1 - \frac{(\underline{a} \cdot \underline{b})^2}{|\underline{a}|^2 |\underline{b}|^2} \right) \\
 &= \frac{1}{4} (|\underline{a}|^2 |\underline{b}|^2 - (\underline{a} \cdot \underline{b})^2) \\
 &= \frac{1}{4} ((\underline{a} \cdot \underline{a})(\underline{b} \cdot \underline{b}) - (\underline{a} \cdot \underline{b})^2) \\
 \therefore A &= \frac{1}{2} \sqrt{(\underline{a} \cdot \underline{a})(\underline{b} \cdot \underline{b}) - (\underline{a} \cdot \underline{b})^2}
 \end{aligned}$$

(d) i. (2 marks)

- ✓ [1] for $\underline{v}$
- ✓ [1] for final answer

$$\begin{aligned}
 \underline{v} &= \left(d \frac{7}{\sqrt{2}} - 10t \right) \\
 \text{When } \frac{7}{\sqrt{2}} - 10t &= 0, \\
 t &= \frac{7}{10\sqrt{2}} = \frac{7\sqrt{2}}{20}
 \end{aligned}$$

ii. (2 marks)

- ✓ [1] for maximum height from the initial position
- ✓ [1] for final answer

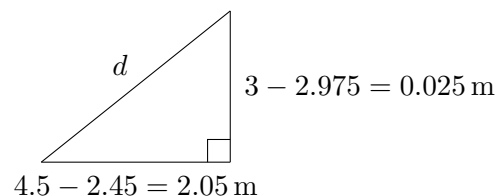
$$\begin{aligned}
 y &= \frac{7}{\sqrt{2}} \times \frac{7\sqrt{2}}{20} - 5 \left(\frac{7\sqrt{2}}{20} \right)^2 = 1.225 \\
 \therefore 1.225 + 1.75 &\approx 2.98 \text{ m}
 \end{aligned}$$

iii. (2 marks)

- ✓ [1] for horizontal distance from the initial position
- ✓ [1] for final answer

$$x = \frac{7}{\sqrt{2}} \times \frac{7\sqrt{2}}{20} = \frac{49}{20} = 2.45 \text{ m}$$

Let d be the distance between the centre of the ball and the centre of the ring.



$$d = \sqrt{2.05^2 + 0.025^2} = 2.05 \text{ m}$$

Question 13

(a) (3 marks)

- ✓ [1] for separation of variables and integration
- ✓ [1] for finding the value of constant
- ✓ [1] for simplifying the equation

$$\begin{aligned}
 \frac{dy}{dx} &= \frac{2y}{x} \\
 \frac{1}{2} \int \frac{1}{y} dy &= \int \frac{1}{x} dx \\
 \frac{1}{2} \ln |y| &= \ln |x| + c \\
 \therefore \text{When } x = 1, y = -1, \\
 c &= 0 \\
 \therefore \frac{1}{2} \ln(-y) &= \ln x \\
 \ln(-y) &= \ln x^2 \\
 -y &= x^2 \\
 y &= -x^2
 \end{aligned}$$

(b) (3 marks)

- ✓ [1] for proving the base case
- ✓ [1] for use of the assumption
- ✓ [1] for final result
- Prove true for $n = 1$:

$$3^3 + 2^3 = 27 + 8 = 35$$

 $\therefore$ true for $n = 1$.

- Assume true for $n = k$

$$3^{3k} + 2^{k+2} = 5M, \text{ where } M \text{ is a positive integer}$$

- Prove true for $n = k + 1$

(d) i. (1 mark)

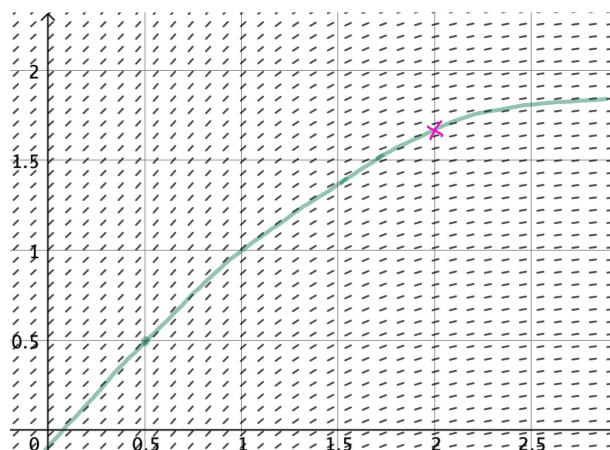
RTP: $3^{3k+3} + 2^{k+3}$ is divisible by 5.

$$\begin{aligned}
 & 3^{3k+3} + 2^{k+3} \\
 &= 3^{3k} \times 3^3 + 2^{k+3} \\
 &= (5M - 2^{k+2}) \times 27 + 2^{k+2} \times 2 \\
 &\dots \text{ from the assumption} \\
 &= 135M - 27 \times 2^{k+2} + 2 \times 2^{k+2} \\
 &= 135M - 25 \times 2^{k+2} \\
 &= 5(27M - 5 \times 2^{k+2}) \\
 &= 5N, \text{ where } N \text{ is a positive integer}
 \end{aligned}$$

$\therefore$ true for $n = k + 1$.

- Conclusion

By mathematical induction, the statement is true for all positive integers.



ii. (1 mark)

✓ [1] for a value in $1.6 \leq y \leq 1.8$

(c) i. (1 mark)

$$\begin{aligned}
 \frac{dT}{dt} &= -k \times (I - A)e^{-kt} \\
 &= -k(T - A)
 \end{aligned}$$

$$y \approx 1.7$$

(e) (3 marks)

$\therefore T = A + (I - A)e^{-kt}$ satisfies the differential equation.

ii. (2 marks)

- ✓ [1] for value of k
- ✓ [1] for final answer

- ✓ [1] for volume when the area under the curve is rotated
- ✓ [1] for volume of cone
- ✓ [1] for final answer

$$I = 1500, A = 20$$

$$T = 20 + 1480e^{-kt}$$

When $t = 5, T = 1200$

$$1200 = 20 + 1480e^{-5k}$$

$$\frac{118}{148} = e^{-5k}$$

$$\therefore k = -\frac{1}{5} \ln \frac{59}{74}$$

When $t = 60$,

$$\begin{aligned}
 T &= 20 + 1480e^{-60k}, \quad \text{where } k = -\frac{1}{5} \ln \frac{59}{74} \\
 &= 118^\circ \text{C (3 sig. fig).}
 \end{aligned}$$

Let V_1 be the volume of the solid when the area under $y = \frac{4\sqrt{2}}{\pi}x$ is rotated and V_2 be the volume of the cone.

$$V_1 = \pi \int_0^{\frac{\pi}{4}} \sec^2 x \, dx$$

$$= \pi [\tan x]_0^{\frac{\pi}{4}}$$

$$= \pi$$

$$V_2 = \frac{1}{3} \times 2\pi \times \frac{\pi}{4}$$

$$= \frac{\pi^2}{6}$$

$$\therefore V = V_1 - V_2$$

$$= \pi - \frac{\pi^2}{6}$$

OR alternatively,

$$\begin{aligned}
 V &= \pi \int_0^{\frac{\pi}{4}} \left(\sec^2 x - \frac{32}{\pi^2} x^2 \right) dx \\
 &= \pi \left[\tan x - \frac{32x^3}{3\pi^2} \right]_0^{\frac{\pi}{4}} \\
 &= \pi \left(1 - \frac{32}{3\pi^2} \times \frac{\pi^3}{64} \right) \\
 &= \pi \left(1 - \frac{\pi}{6} \right) \\
 &= \pi - \frac{\pi^2}{6}
 \end{aligned}$$

Question 14

(a) (4 marks)

- ✓ [1] for changing limits
- ✓ [1] for expressing the integral in terms of u
- ✓ [1] for integrating correctly
- ✓ [1] for final answer

$$\begin{aligned}
 &\int_1^9 \frac{\sqrt{x} + 2}{\sqrt{x}\sqrt{1 + \sqrt{x}}} dx \\
 &\left| \begin{array}{ll} u = 1 + \sqrt{x} & \sqrt{x} = u - 1 \\ du = \frac{1}{2\sqrt{x}} dx & dx = 2\sqrt{x} du \end{array} \right.
 \end{aligned}$$

When $x = 9$, $u = 4$

When $x = 1$, $u = 2$

$$\begin{aligned}
 &= \int_2^4 \frac{u + 1}{(u - 1)\sqrt{u}} \times 2(u - 1) du \\
 &= 2 \int_2^4 u^{\frac{1}{2}} + u^{-\frac{1}{2}} du \\
 &= 2 \left[\frac{2}{3} u^{\frac{3}{2}} + 2u^{\frac{1}{2}} \right]_2^4 \\
 &= 2 \left(\frac{2}{3} \times 8 + 2 \times 2 - \frac{2}{3} \times 2\sqrt{2} - 2\sqrt{2} \right) \\
 &= 2 \left(\frac{28}{3} - \frac{10}{3} \sqrt{2} \right) \\
 &= \frac{56}{3} - \frac{20}{3} \sqrt{2}
 \end{aligned}$$

(b) i. (2 marks)

- ✓ [1] for domain
- ✓ [1] for range

Domain:

$$-1 \leq h - \frac{\sqrt{3}}{2} \leq 1$$

$$-1 + \frac{\sqrt{3}}{2} \leq h \leq 1 + \frac{\sqrt{3}}{2}$$

However, since $h \geq 0$,

$$0 \leq h \leq 1 + \frac{\sqrt{3}}{2}$$

Range:

$$\because 0 \leq \cos^{-1} \left(h - \frac{\sqrt{3}}{2} \right) \leq \pi$$

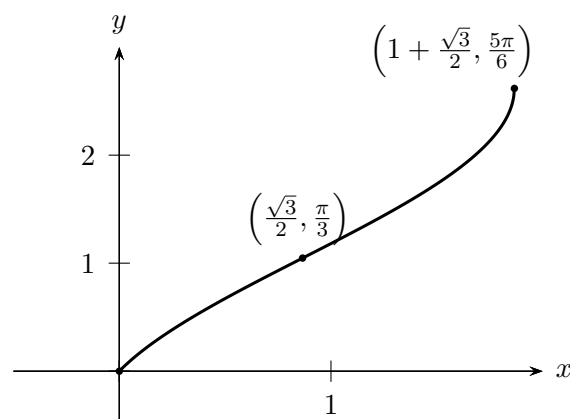
$$-\frac{\pi}{6} \leq \frac{5\pi}{6} - \cos^{-1} \left(h - \frac{\sqrt{3}}{2} \right) \leq \frac{5\pi}{6}$$

However, since $V \geq 0$

$$0 \leq \frac{5\pi}{6} - \cos^{-1} \left(h - \frac{\sqrt{3}}{2} \right) \leq \frac{5\pi}{6}$$

ii. (3 marks)

- ✓ [1] for shape
- ✓ [1] for endpoints
- ✓ [1] for point of inflexion



iii. (1 mark)

$$\because 50 \text{ L} = \frac{1}{20} \text{ m}^3,$$

$$V = \frac{1}{20}t \text{ or } V = 0.05t$$

iv. (1 mark) When $V = \frac{5\pi}{6}$,

$$\frac{5\pi}{6} = \frac{1}{20}t$$

$$\therefore t = \frac{5\pi}{6} \times 20$$

$$t = \frac{50\pi}{3}$$

v. (3 marks)

✓ [1] for $\frac{dV}{dh}$

✓ [1] for use of chain rule

✓ [1] for final answer

$$\frac{dV}{dh} = \frac{1}{\sqrt{1 - \left(h - \frac{\sqrt{3}}{2}\right)^2}}$$

$$\begin{aligned} \frac{dh}{dt} &= \frac{dh}{dV} \times \frac{dV}{dt} \\ &= \sqrt{1 - \left(h - \frac{\sqrt{3}}{2}\right)^2} \times \frac{1}{20} \end{aligned}$$

When $h = \sqrt{3}$,

$$\begin{aligned} \frac{dh}{dt} &= \sqrt{1 - \left(\frac{\sqrt{3}}{2}\right)^2} \times \frac{1}{20} = \frac{1}{2} \times \frac{1}{20} \\ &= \frac{1}{40} \end{aligned}$$

$\therefore \frac{1}{40}$ or 0.025 m/min.